OFFLINE Map construction

No original mapping images



Render

rendering viewpoints



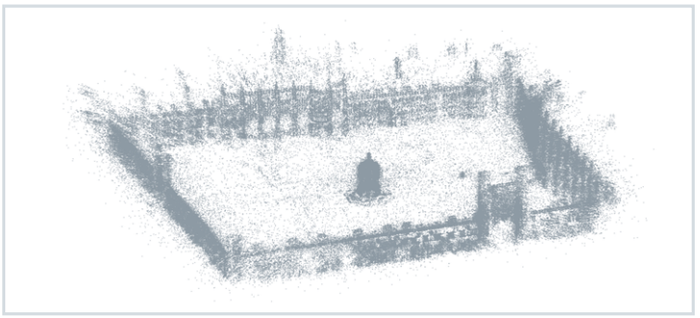
Prebuilt Gaussian scene

2D or 3D Gaussians

Rendered observations

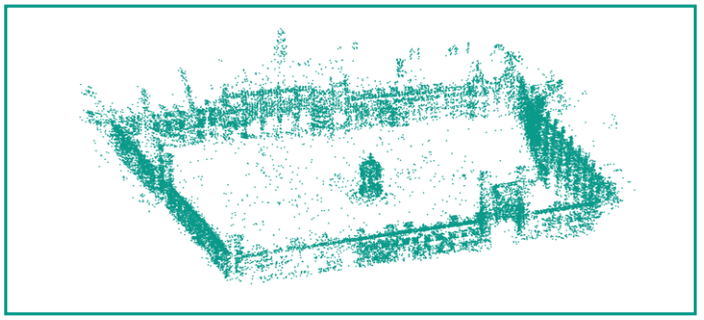
RGB · opacity · depth

1 Track-based reconstruction



2

Localization-aware selection



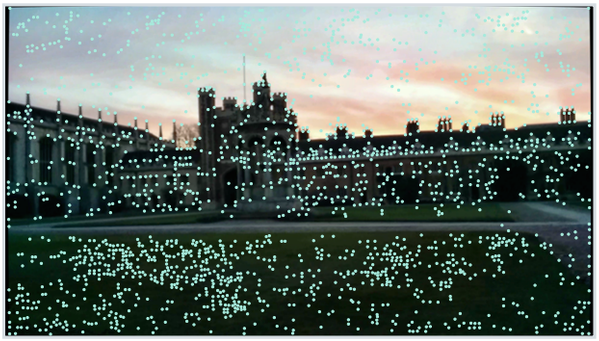
Full map

201,063 landmarks

Compact map

44,631 · points + descriptors

ONLINE Single-stage localization



Global 2D–3D matching

Robust pose



Query features

No rendering

6-DoF pose